

Jan Tolsdorf

DR. RER. NAT. (EQUIVALENT TO PH.D.) · HE/HIM

Bochum, Germany

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Research interests: human-centered privacy & security, trustworthy AI, human factors research

Education

Ph.D. in Computer Science (Dr. rer. nat., summa cum laude)

GEORG-AUGUST-UNIVERSITÄT GÖTTINGEN

Advisors: Professors Dr.-Ing. Delphine Reinhardt and Dr.-Ing. Luigi Lo Iacono

07/2019 - 08/2022

Göttingen, Germany

M.Sc. in Media Technology

TECHNISCHE HOCHSCHULE KÖLN UNIVERSITY OF APPLIED SCIENCES

Thesis Advisors: Professors Dr.-Ing. Luigi Lo Iacono and Dr.-Ing. Matthew Smith

09/2016 - 01/2019

Cologne, Germany

B.Sc. in Media Technology

TECHNISCHE HOCHSCHULE KÖLN UNIVERSITY OF APPLIED SCIENCES

Thesis Advisors: Professors Dr.-Ing. Luigi Lo Iacono and Dr.-Ing. Arnulph Fuhrmann

09/2012 - 08/2016

Cologne, Germany

Professional Experience

Postdoctoral Researcher

MAX PLANCK INSTITUTE FOR SECURITY AND PRIVACY (MPI-SP)

Security and Privacy Engineering Lab (SPRING)

Supervisor: Prof. Carmela Troncoso, Ph.D.

11/2025 - present

Bochum, Germany

Postdoctoral Associate (AI Fellow)

THE GEORGE WASHINGTON UNIVERSITY

Institute for Trustworthy AI in Law & Society

Department of Computer Science

Supervisor: Prof. Adam J. Aviv, Ph.D.

02/2024 - 09/2025

Washington D.C., USA

Part-Time Faculty

THE GEORGE WASHINGTON UNIVERSITY

Department of Computer Science

08/2024 - 12/2024

Washington D.C., USA

External Lecturer

HOCHSCHULE BONN-RHEIN-SIEG UNIVERSITY OF APPLIED SCIENCES

Institute for Cyber Security & Privacy

Department of Computer Science

03/2023 - 09/2023

Sankt Augustin, Germany

Research Associate

HOCHSCHULE BONN-RHEIN-SIEG UNIVERSITY OF APPLIED SCIENCES

Institute for Cyber Security & Privacy

Department of Computer Science

Supervisor: Prof. Dr.-Ing. Luigi Lo Iacono

06/2020 - 02/2024

Sankt Augustin, Germany

Research Associate

TECHNISCHE HOCHSCHULE KÖLN UNIVERSITY OF APPLIED SCIENCES

Institute for Media and Photo Technology

Faculty of Information, Media, and Electrical Engineering

Supervisor: Prof. Dr.-Ing. Luigi Lo Iacono

03/2018 - 05/2020

Cologne, Germany

Student Assistant

G&L GEIßENDÖRFER UND LESCHINSKY GMBH

Team Software Development

12/2015 - 02/2018

Cologne, Germany

Publications

PEER-REVIEWED RESEARCH ARTICLES

- [CCS 26] Junho Eum, **Jan Tolsdorf**, Adryana Hutchinson, Smirity Kaushik, and Adam J. Aviv. “User Perceptions of Online Age Verification Systems in the US”. In: *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. Accepted for publication. 2026.
- [SP 26] David Langer, **Jan Tolsdorf**, and Luigi Lo Iacono. “LISA: A Scale-Optimized and Psychometrically-Validated Instrument for the Lightweight Assessment of Organizational Information Security Awareness in Heterogeneous Organizations”. In: *Proceedings of the 47th IEEE Symposium on Security and Privacy (IEEE SP)*. 2026.
- [FAccT 26] **Jan Tolsdorf**, Alan F. Luo, Monica Kodwani, Junho Eum, Mahmood Sharif, Michelle L. Mazurek, and Adam J. Aviv. “How Probing for Problems and Bias Affects Perceptions of AI Chatbot Trustworthiness”. In: *Proceedings of the 9th ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2026.
- [IWPE 25] Florian Dehling, **Jan Tolsdorf**, and Luigi Lo Iacono. “Right Here, Right Now: User Perceptions of In-Place Contextual Privacy Options”. In: *International Workshop on Privacy Engineering (IWPE)*. 2025.
- [CCS 25] **Jan Tolsdorf**, David Langer, and Luigi Lo Iacono. “Phishing Susceptibility and the (In-)Effectiveness of Common Anti-Phishing Interventions in a Large University Hospital”. In: *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 2025.
- [ConPro 25] **Jan Tolsdorf**, Alan F. Luo, Monica Kodwani, Junho Eum, Mahmood Sharif, Michelle L. Mazurek, and Adam J. Aviv. “On a Scale of 1 to 5, How Reliable Are AI User Studies? A Call for Developing Validated, Meaningful Scales and Metrics about User Perceptions of AI Systems”. In: *IEEE SPW Workshop on Technology and Consumer Protection (ConPro)*. 2025.
- [SOUPS 25] **Jan Tolsdorf**, Alan F. Luo, Monica Kodwani, Junho Eum, Mahmood Sharif, Michelle L. Mazurek, and Adam J. Aviv. “Safety Perceptions of Generative AI Conversational Agents: Uncovering Perceptual Differences in Trust, Risk, and Fairness”. In: *Proceedings of the 21st Symposium on Usable Privacy and Security (SOUPS)*. 2025.
- [PETS 24] Florian Dehling, **Jan Tolsdorf**, Hannes Federrath, and Luigi Lo Iacono. “Internet Users’ Willingness to Disclose Biometric Data for Continuous Online Account Protection: An Empirical Investigation”. In: *Proceedings on Privacy Enhancing Technologies (PoPETs) 2024.2* (2024).
- [HCICPT 24] **Jan Tolsdorf** and Luigi Lo Iacono. “Expert Perspectives on Information Security Awareness Programs in Medical Care Institutions in Germany”. In: *Proceedings of the 6th International Conference on HCI for Cybersecurity, Privacy and Trust (HCI-CPT)*. 2024.
- [BIT 22] **Jan Tolsdorf**, Florian Dehling, and Luigi Lo Iacono. “Data Cart – Designing a Tool for the GDPR-compliant Handling of Personal Data by Employees”. In: *Behaviour & Information Technology (BIT)* 41.10 (2022). IF: 3.7.
- [PETS 22] **Jan Tolsdorf**, Delphine Reinhardt, and Luigi Lo Iacono. “Employees’ privacy perceptions: exploring the dimensionality and antecedents of personal data sensitivity and willingness to disclose”. In: *Proceedings on Privacy Enhancing Technologies (PoPETs) 2022.2* (2022).
- [SPOSE 22] Stephan Wiefeling, **Jan Tolsdorf**, and Luigi Lo Iacono. “Data Protection Officers’ Perspectives on Privacy Challenges in Digital Ecosystems”. In: *Proceedings of the 4th Workshop on Security, Privacy, Organizations, and Systems Engineering (SPOSE)*. 2022.

- [TrustBus 21] Florian Dehling, Denis Feth, Svenja Polst, Bianca Steffes, and **Jan Tolsdorf**. “Components and Architecture for the Implementation of Technology-driven Employee Data Protection”. In: *Proceedings of the 18th International Conference on Trust, Privacy and Security in Digital Business (TrustBus)*. 2021.
- [DuD 21b] Svenja Polst, **Jan Tolsdorf**, Florian Dehling, and Denis Feth. “Verarbeitung von Beschäftigtendaten”. In: *Datenschutz und Datensicherheit (DuD)* 45.1 (2021).
- [DuD 21a] **Jan Tolsdorf**, Florian Dehling, and Denis Feth. “Benutzerfreundlicher Datenschutz in Cloud-basierten Office-Paketen”. In: *Datenschutz und Datensicherheit (DuD)* 45.1 (2021).
- [PETS 21] **Jan Tolsdorf**, Florian Dehling, Delphine Reinhardt, and Luigi Lo Iacono. “Exploring mental models of the right to informational self-determination of office workers in Germany”. In: *Proceedings on Privacy Enhancing Technologies (PoPETs)* 2021.3 (2021).
- [APF 21] **Jan Tolsdorf**, Michael Fischer, and Luigi Lo Iacono. “A Case Study on the Implementation of the Right of Access in Privacy Dashboards”. In: *Proceedings of the 9th Annual Privacy Forum (APF)*. 2021.
- [IWPE 21] Stephan Wiefeling, **Jan Tolsdorf**, and Luigi Lo Iacono. “Privacy Considerations for Risk-based Authentication Systems”. In: *Proceedings of the 7th IEEE International Workshop on Privacy Engineering (IWPE)*. 2021.
- [IRIS 20] Christian K. Bosse, Aljoscha Dietrich, Patricia Kelbert, Hagen Küchler, Hartmut Schmitt, **Jan Tolsdorf**, and Andreas Weißner. “Beschäftigtendatenschutz: Rechtliche Anforderungen und Technische Lösungskonzepte”. In: *Tagungsband des 23. Internationalen Rechtsinformatik Symposions IRIS 2020*. 2020.
- [DuD 20] **Jan Tolsdorf**, Christian K. Bosse, Aljoscha Dietrich, Denis Feth, and Hartmut Schmitt. “Privatheit am Arbeitsplatz - Transparenz und Selbstbestimmung bei Arbeit 4.0”. In: *Datenschutz und Datensicherheit (DuD)* 44.3 (2020).
- [AsiaUSEC 20] **Jan Tolsdorf** and Florian Dehling. “In Our Employer We Trust: Mental Models of Office Workers’ Privacy Perceptions”. In: *Proceedings of the 1st Asian Workshop on Usable Security (AsiaUSEC, FC workshop)*. 2020.
- [ITG News 20] **Jan Tolsdorf**, Florian Dehling, and Luigi Lo Iacono. “Take Back Control! The Use of Mental Models to Develop Privacy Dashboards”. In: *ITG News* 8.3 (2020). Edited by Informationstechnische Gesellschaft im VDE.
- [EuroUSEC 20] **Jan Tolsdorf** and Luigi Lo Iacono. “Vision: Shred If Insecure – Persuasive Message Design as a Lesson and Alternative to Previous Approaches to Usable Secure Email Interfaces”. In: *Proceedings of the 5th European Workshop on Usable Security (EuroUSEC)*. 2020.
- [ECCWS 18] Nils Gruschka, Luigi Lo Iacono, and **Jan Tolsdorf**. “Classification of Android App Permissions: Tell Me What App You Are and I Tell You What You Are Allowed to Do”. In: *Proceedings of the 17th European Conference on Cyber Warfare and Security (ECCWS)*. 2018.
- [TrustBus 17] Hoai Viet Nguyen, **Jan Tolsdorf**, and Luigi Lo Iacono. “On the Security Expressiveness of REST-Based API Definition Languages”. In: *Proceedings of the 14th International Conference on Trust, Privacy and Security in Digital Business (TrustBus)*. 2017.

FURTHER PUBLICATIONS AND REPORTS

- [BMG 25] Luigi Lo Iacono, **Jan Tolsdorf**, and David Langer. *Abschlussbericht zum Projekt MedISA: Medical Centre Employee Centered Information Security Awareness*. Technical report. Hochschule Bonn-Rhein-Sieg, 2025.
- [KTM 23b] **Jan Tolsdorf** and Luigi Lo Iacono. “Faktor Mensch: Risiko und Chance: Informationssicherheitsbewusstsein in medizinischen Einrichtungen – Expertenansichten (Teil 2)”. In: *Krankenhaus Technik+Management (KTM)* 2023.9 (2023).
- [KTM 23a] **Jan Tolsdorf** and Luigi Lo Iacono. “Medizinisches Personal sensibilisieren: Bewusstsein für Gefährdungen der Patientenversorgung aus dem Cyberraum schaffen – Teil 1”. In: *Krankenhaus Technik+Management (KTM)* 2023.7-8 (2023).
- [BMBF 22] Luigi Lo Iacono and **Jan Tolsdorf**. *Schlussbericht des Teilvorhabens “Konzeptionierung, Implementierung und Evaluation von Privacy-Dashboards im Arbeitnehmerdatenschutz” im Rahmen des Verbundprojekts “TrUSD - Transparente und selbstbestimmte Ausgestaltung der Datennutzung im Unternehmen”*. Technical report. Hochschule Bonn-Rhein-Sieg, 2022.
- [DISS 22] **Jan Tolsdorf**. “Investigation of Information Privacy in Employment: Fundamental Knowledge and Practical Solutions for the Human-Centered Design of Measures to Preserve the Right to Informational Self-Determination in Employment”. Dissertation. 2022.

BOOK CHAPTERS

- [Book 23b] Eduard C. Groen, Denis Feth, Svenja Polst, **Jan Tolsdorf**, Stephan Wiefeling, Luigi Lo Iacono, and Hartmut Schmitt. "Achieving Usable Security and Privacy Through Human-Centered Design". In: *Human Factors in Privacy Research*. Edited by Nina Gerber, Alina Stöver, and Karola Marky. 2023.
- [GI 23] **Jan Tolsdorf**. "Informationelle Privatheit im Beschäftigungsverhältnis". In: *Ausgezeichnete Informatikdisertationen 2022 (Band D23)*. 2023.
- [Book 23a] **Jan Tolsdorf** and Luigi Lo Iacono. "Data Cart: A Privacy Pattern for Personal Data Management in Organizations". In: *Human Factors in Privacy Research*. Edited by Nina Gerber, Alina Stöver, and Karola Marky. 2023.

3rd Party Funded Research Projects

I have worked on the following third-party funded research projects:

How Users Perceive and Expose Bias in AI Systems

08/2024 - 07/2025

FUNDED BY THE INSTITUTE FOR TRUSTWORTHY AI IN LAW & SOCIETY (TRAILS)

D.C., USA

Investigating lay users' perceptions of AI chatbot trustworthiness.

PI: Professors Adam J. Aviv, PhD and Michelle L. Mazurek, PhD

MedISA – "Medical Centre Employee Centered Information Security Awareness"

12/2021 - 02/2024

FUNDED BY THE GERMAN FEDERAL MINISTRY OF HEALTH (BMG)

Germany

Investigating and documenting measures to increase information security awareness in medical care facilities.

PI: Prof. Dr.-Ing. Luigi Lo Iacono

D'ACCORD

09/2021 - 11/2021

FUNDED BY THE GERMAN FEDERAL MINISTRY OF EDUCATION AND RESEARCH (BMBF)

Germany

Investigating human factors and development of privacy cockpits for digital ecosystems.

PI: Prof. Dr.-Ing. Luigi Lo Iacono

TrUSD – "Transparent and Self-determined Design of Data Use in Companies"

01/2019 - 08/2021

FUNDED BY THE GERMAN FEDERAL MINISTRY OF EDUCATION AND RESEARCH (BMBF)

Germany

Investigating human factors in employee privacy and the development of privacy dashboards to increase transparency and self-determination in employment.

PI: Prof. Dr.-Ing. Luigi Lo Iacono

KieBox

03/2018 - 01/2019

FUNDED BY THE GERMAN FEDERAL MINISTRY FOR ECONOMIC AFFAIRS AND ENERGY (MWi)

Germany

Development of a usable secure e-mail solution in cooperation with a private company.

PI: Prof. Dr.-Ing. Luigi Lo Iacono

Media Coverage

Heise Online, "Study: [EXTERNAL] tags do not protect against phishing",

2025

<https://www.heise.de/news/Studie-EXTERN-Tags-schuetzen-nicht-vor-Phishing-11092509.html>

Golem, "[EXTERN]-Tags schützen kaum vor Datenklau via Phishing", <https://www.golem.de/news/studie-zeigt-extern-tags-schuetzen-kaum-vor-datenklau-via-phishing-2511-202649.html>

2025

Krankenhaus IT, "Studie zeigt: Viele Anti-Phishing-Maßnahmen schützen Kliniken nur unzureichend",

2025

<https://www.krankenhaus-it.de/item.4689/studie-zeigt-viele-anti-phishing-massnahmen-schuetzen-kliniken-nur-unzureichend.html>

Frankfurter Allgemeine Zeitung (FAZ) , “Simulation zeigt Risiken: Große Gefahr durch Phishing in Kliniken”, https://www.faz.net/aktuell/rhein-main/region-und-hessen/gefahr-durch-phishing-in-kliniken-accg-200266968.html	2025
Deutsches Ärzteblatt , “Cybersicherheit: Angriffe leicht möglich”, https://www.aerzteblatt.de/archiv/titel/dae/2025/25/cybersicherheit-angriffe-leicht-moeglich-693174cf-e8ae-4ffe-ba9d-542e799a5b36	2025

Awards, Fellowships, & Grants

Co-Initiator of “HuSecMe” under the StartUpSecure Initiative , German Federal Ministry of Research, Technology and Space	€ 580,000	2025
Top Reviewers Award , ACM CCS		2025
Nominee for the Dissertation Award 2022 of the German Informatics Society , Georg-August-Universität Göttingen		2023
Doctoral Thesis Award Bonn-Rhein-Sieg University Society 2022 , Hochschulgesellschaft Bonn-Rhein-Sieg e. V.	€ 1,000	2023
GAUSS Book Award in recognition of outstanding doctoral achievements , Georg-August-Universität Göttingen		2022

Invited Talks & Guest Lectures

Invited talk , GCI Seminar Series, Rochester Institute of Technology, Rochester, New York, USA “Enabling Human-Centered Privacy and Security Engineering”	Fall ‘24
Guest lecture , Barnard College, Columbia University, New York, USA “Conceptualizations of Privacy”	Fall ‘24
Invited talk , The George Washington University, Washington, D.C., USA “Enabling Human-Centered Privacy and Security Engineering”	Spring ‘23
Invited talk , Stammtisch Cyber Security Cluster Bonn e. V., Bonn, Germany “Das Verfahren ist nicht neu [...] aber Sie erreichen damit niemanden mehr’ – Maßnahmen zur Steigerung von Informationssicherheitsbewusstsein in medizinischen Versorgungseinrichtungen. (Engl.: Measures to increase information security awareness in medical care facilities.)”	Winter ‘23
Invited talk , Hochschulgesellschaft Bonn-Rhein-Sieg e. V., Sankt Augustin, Germany “Employee-Centered Design of Technical and Organizational Privacy Measures”	Winter ‘22
Invited talk , CS Colloquium, Karlstad University, Karlstad, Sweden “Employee-Centered Design of Technical and Organizational Privacy Measures”	Winter ‘22
Invited talk , AFSecurity Seminar, University of Oslo, Oslo, Norway “Employee-Centered Design of Technical and Organizational Privacy Measures”	Winter ‘22
Guest lecture , Friedrich-Alexander-Universität, Erlangen-Nürnberg, Germany “Employees’ Mental Models of the Right to Privacy: Insights from Germany”	Summer ‘22
Invited talk , Usable Security and Privacy (USP) Day, VDE ITG, Online, Germany “Data Cart - Designing a tool for the GDPR-compliant handling of personal data by employees”	Summer ‘22
Guest lecture , Georg-August-Universität, Göttingen, Germany “Employees’ Mental Models of the Right to Privacy: Insights from Germany”	Winter ‘21

Guest lecture , Georg-August-Universität, Göttingen, Germany <i>“Employees’ Mental Models of the Right to Privacy: Insights from Germany”</i>	Winter ‘20
Invited talk , Georg-August-Universität Göttingen, Göttingen, Germany <i>“Das Privacy-paradoxon – Der Scheinbare Widerspruch Zwischen Bedenken Und Verhalten Beim Datenschutz (Engl.: The Privacy Paradox - the Apparent Contradiction Between Privacy Concerns and Behavior)”</i>	Spring ‘19

Teaching & Mentoring

LECTURER/ INSTRUCTOR

Co-Instructor / External Lecturer in “ Empirical Security Research ”	<i>M.Sc. course</i>	Winter ‘26
Co-Instructor in “ CSCI 4533/6533 Introduction to Usable Security and Privacy ”	<i>BS, MS, PhD course</i>	Fall ‘24
Instructor / External Lecturer in “ Privacy Engineering ”	<i>B.Sc. Course</i>	Summer ‘23

TUTOR & TEACHING ASSISTANCE

Teaching Assistant in “ Privacy and Security ” Supervising and evaluating seminar papers	<i>B.Sc. Course</i>	Winter ‘23/‘24 Winter ‘22/‘23
Teaching Assistant in “ Data Protection, IT Law and Privacy ” Preparing courses and exercises	<i>B.Sc. Course</i>	Winter ‘21/‘22 Winter ‘20/‘21
Teaching Assistant in “ Advanced Topics in Security and Privacy ” Supervising and evaluating seminar papers	<i>M.Sc. Course</i>	Summer ‘21
Teaching Assistant in “ Privacy and Security ” Supervising and evaluating seminar papers	<i>M.Sc. Course</i>	Summer ‘20 Winter ‘19/‘20
Teaching Assistant in “ Web Programming ” Supervising lab exercises	<i>B.Sc. Course</i>	Summer ‘20 Summer ‘19 Summer ‘18
Teaching Assistant in “ Informatics 1 ” Tutoring and reviewing programming exercises	<i>B.Sc. Course</i>	Winter ‘14/‘15

STUDENT MENTORING

Arwa Alshadi, GWU, Ph.D. student
Junho Eum, GWU, Ph.D. student
Wellington Esposito Barbosa, GWU, Ph.D. student
Monica Kodwani, GWU, Ph.D. student
Yanzi Lin, CMU, Ph.D. student and intern at MPI-SP
Joseph Staller, GWU, BS student, Alumni
Philip Erdmann, H-BRS, BS student, Alumni
Marian Hönscheid, H-BRS, BS student, Alumni
Cole Knie, GWU, MS student, Alumni

SUPERVISED STUDENT THESES

S. Gerigk. “Berechtigte Bedenken oder Panikmache? Eine datengetriebene Analyse öffentlicher Kritik an der European Digital Identity Wallet und ihrer technischen Berechtigung (Engl.: Valid Concerns or Alarmism? A Data-Driven Analysis of Public Criticism of the European Digital Identity Wallet and Its Technical Viability.)”	<i>M.Sc. Thesis</i>	In progress
A. Satgunanathan. “When Users Have No Choice: An Experimental Study of Trust and Privacy in Mandatory Human-AI Service Interactions”	<i>M.Sc. Thesis</i>	In progress
P. Klein. “Wie Nutzende mit Fremdprivatheit in KI-Anwendungen umgehen (Engl.: How users deal with other people’s privacy in AI applications.)”	<i>B.Sc. Thesis</i>	2025
A. Portz. “Datenschutzkonformität von Dashcams. (Engl.: Data Protection Compliance of Dashcams.)”	<i>B.Sc. Thesis</i>	2024
L. Rohé. “Auswirkung von Nudging auf Security Fatigue am Beispiel Passwortwahl. (Engl.: Effects of Nudging on Security Fatigue Using the Example of Password Selection.)”	<i>B.Sc. Thesis</i>	2022
M. Fischer. “Privacy Dashboards – Untersuchung verschiedener Dashboards von Unternehmen im Hinblick auf die Anforderungen der DSGVO. (Engl.: Privacy Dashboards - Examination of Various Dashboards of Companies Regarding the Requirements of the GDPR.) ”	<i>B.Sc. Thesis</i>	2021
F. Dehling. “Investigations on Utilizing Web Browser Caches for User Tracking.”	<i>M.Sc. Thesis</i>	2020

Service

POSTER CO-CHAIR

Privacy Enhancing Technologies Symposium: 2026

PROGRAM COMMITTEE MEMBERSHIP

Annual Privacy Forum: 2023, 2024, 2025

ACM CCS: 2025, 2026

EuroUSEC: 2023, 2024

Usable Security and Privacy Workshop: 2020, 2021, 2022, 2023, 2024, 2025

Privacy Enhancing Technologies Symposium: 2025, 2026, 2027

EXT. REVIEWER

ACM CHI: 2022, 2025, 2026

International Journal of Human-Computer Interaction: 2022, 2023, 2024

Behaviour & Information Technology: 2022

EXAMINER

Bachelor’s Thesis, Bonn-Rhein-Sieg University of Applied Sciences, 2025

Bachelor’s Thesis, Bonn-Rhein-Sieg University of Applied Sciences, 2024

Master’s Thesis, University of Oslo, 2021

ORGANIZATIONAL

Member of the MPI-SP Ethics Review Board, 2026

Co-organizer of Spar(k)lingScience at MPI-SP, 2025, 2026